METHOD STATEMENT

Freezo Coat One — Flexible Polyurethane Floor System for Cold Rooms

Project Name		**Document No.**	OMX-SMS-03
Main Contractor		**Revision**	01
Consultant		**Date of Issue**	Nov 2026
System	Freezo Coat One	**Status**	For Approval

1. Purpose

This method statement defines the sequence, resources, inspection and test plan and safety measures for the installation of the Freezo Coat One system on site. It describes how each layer of the system is prepared, mixed, applied, inspected and protected so that the completed system achieves the performance stated in the O MIX Technical Data Sheets of its components.

2. Scope of Work

Freezo Coat One is a flexible floor and wall system for cold rooms and freezers. A solvent-free epoxy primer and epoxy filler prepare the substrate, and a solvent-free, highly flexible polyurethane top coat resists thermal shock, abrasion and chemicals from -35 °C to +150 °C.

- Substrate inspection, testing and acceptance.
- Surface preparation and repair of defects.
- Stage 1: Primer — O-Prime SF.
- Stage 2: Filling / levelling layer — O-Putty.
- Stage 3: Flexible PU top coat — Freezo Coat.
- Curing, protection, final testing and handover.

2.1 System build-up

Stage	Product	Function	Consumption
1	O-Prime SF	Primer	0.10 – 0.15 kg/m²
2	O-Putty	Filling / levelling layer	0.25 – 0.30 kg/m²
3	Freezo Coat	Flexible PU top coat	0.5 kg/m²

2.2 Typical areas of use

- Cold rooms and freezers.
- Food processing plants.
- Pharmaceutical cold stores.
- Areas with thermal cycling.

3. Definitions

Term	Meaning
System	All layers of the build-up applied in the stated sequence.
Hold point	Stage at which work stops until QA/QC (and the consultant, where required) approve.
Recoat window	Minimum and maximum time between two layers for proper inter-layer bonding.
Dew point	Temperature at which moisture condenses; the substrate must be at least 3 °C above it.
Mock-up / trial area	Area applied before the main works to approve materials, method and appearance.
Pot life	Time a mixed multi-component material remains workable.
WFT / DFT	Wet film thickness during application / dry film thickness after cure.

4. Roles & Responsibilities

Role	Responsibility
Project Manager	Planning, resources, approvals and coordination with the consultant and main contractor.
Site Engineer	Daily supervision; confirms substrate readiness, climate, mixing, consumption and recoat windows; keeps daily records.
QA/QC Inspector	Carries out tests at each hold point, records results and signs the inspection checklist.
HSE Officer	Toolbox talks, PPE, ventilation, safe access and storage; stops unsafe work.
Applicator Foreman	Leads the application team and follows this method statement; reports deviations.
O MIX Technical Representative	Technical support, site visits and advice on product use when requested.

5. Materials

Product	Description	Packaging
O-Prime SF	Solvent-Free Epoxy Primer & Sealer for Concrete Floors	5 kg sets: Part A in pail + Part B in jerrycan.
O-Putty	Two-Component Epoxy Filler for Concrete, Steel & Equipment	5 kg sets (Part A + Part B), metal tins.
Freezo Coat	Solvent-Free Two-Component PU Coating for Cold Rooms	20 kg sets (15 kg A + 5 kg B) and 5 kg sets (3.75 kg + 1.25 kg).

5.1 Key product data

Product	Mixing ratio	Pot life / working time	Recoat / drying
O-Prime SF	As recommended — pre-weighed sets	40 min	10 hours (max. 24 hours)
O-Putty	1 : 1 by volume	60 min	24 hours
Freezo Coat	3 : 1 by weight	40 – 60 min	Full cure before cooling the room

6. Tools & Equipment

- Diamond grinder and/or shot blaster with dust extraction; industrial vacuum
- Slow-speed drill mixer (300 – 500 rpm) with spiral paddle
- Rollers, brushes, squeegees, notched trowels and steel trowels as required by each stage
- Scales and measuring containers for part mixing (where permitted)
- Thermo-hygrometer, surface thermometer and dew-point calculator
- Moisture meter and plastic sheets for moisture testing
- Masking tape, protective sheeting, barriers and warning signs
- Lighting and extension cables with residual current protection
- Wet-film comb gauge
- Pull-off adhesion tester
- Spiked roller and spiked shoes

6.1 Personal protective equipment

- Safety shoes, helmet and high-visibility vest
- Chemical-resistant gloves (nitrile) and safety goggles
- Coveralls with long sleeves
- Dust masks (P2) during grinding and powder mixing
- Respirators with organic vapour cartridges for solvent-containing products and confined spaces
- Knee pads for floor work

7. Storage & Handling

- Store all products in original, unopened packaging in a dry, shaded and ventilated store at the temperatures stated in each TDS.
- Keep Part A, Part B (and Part C) of each kit together; do not mix components from different batches.
- Protect liquids from frost and direct sun, and powders from moisture.
- Store solvent-containing products away from heat, sparks and ignition sources.
- Use materials on a first-in, first-out basis and record batch numbers for each area.
- Condition materials to +15 °C to +25 °C for 24 hours before use in hot or cold weather.

8. Pre-Start Requirements

- Approved material submittal, this method statement and the inspection and test plan.
- Approved drawings showing areas, joints, upstands, drains and details.
- Work area handed over clean, free of other trades, with safe access, lighting and power.
- Mock-up / trial area executed and approved for system, colour and finish.
- Toolbox talk held with the team on this method statement and HSE measures.
- Climate monitoring equipment calibrated and available.

9. Substrate Inspection & Acceptance (Hold Point)

Check	Acceptance criteria	Method
Concrete age	≥ 28 days (cement-based substrates)	Site records
Compressive strength	≥ 25 N/mm²	Rebound hammer (ASTM C805) or cores
Surface tensile (pull-off) strength	≥ 1.5 N/mm²	Pull-off test ASTM D7234
Moisture content	< 4 % (no moisture under plastic sheet after 16 h)	Moisture meter; plastic sheet test ASTM D4263
Surface profile	CSP 2 – 3 (primers / thin coats), CSP 3 – 5 (self-levelling / thick layers)	ICRI 310.2R comparator chips
Flatness	As per project specification	3 m straight edge
Climate	Substrate ≥ 3 °C above dew point; RH ≤ 85 %	ISO 8502-4 / thermo-hygrometer

10. Surface Preparation

1. Remove laitance, curing compounds, oil, paint and loose material by shot blasting or diamond grinding to the required profile.
2. Open cracks to a V-groove and remove dust; repair holes, voids and spalls.
3. Grind high spots and remove sharp edges; form 45° coves at wall / floor junctions where required.
4. Vacuum all dust with an industrial vacuum immediately before priming.
5. Mask and protect adjacent surfaces, drains and fixtures.
6. Record substrate test results and obtain QA/QC approval before priming (hold point).

11. Application Sequence

Apply the system stage by stage in the order below. Each stage must be inspected and approved before the next stage starts, and the next stage must be applied within the recoat window stated.

11.1 Stage 1 — Primer: O-Prime SF

Consumption: 0.10 – 0.15 kg/m². Time before next stage: Min. 10 h, max. 24 h.

1. Check batch numbers and condition the kit to +15 °C to +25 °C.
2. Pour all of Part B into Part A.
3. Mix 2 – 3 minutes below 300 rpm until uniform.
4. Mix complete sets only; do not thin.
5. Apply by roller at 0.10 – 0.15 kg/m² without puddles.
6. Apply a second coat on very porous substrates.
7. Overcoat with the next system layer within 24 hours, or broadcast 0.1 – 0.4 mm quartz sand into the wet primer.

11.2 Stage 2 — Filling / levelling layer: O-Putty

Consumption: 0.25 – 0.30 kg/m². Time before next stage: 24 h, then light sanding.

1. Check batch numbers and condition the kit to +15 °C to +25 °C.
2. Take equal volumes of Part A and Part B.
3. Mix on a clean board until uniform in colour, with no streaks.
4. Press the filler firmly into cracks and voids with a steel spatula.
5. Skim the surface at 0.25 – 0.30 kg/m² where used as a filling layer.
6. After 24 hours, sand lightly and remove dust before overcoating.

11.3 Stage 3 — Flexible PU top coat: Freezo Coat

Consumption: 0.5 kg/m². Time before next stage: Full cure before cooling the room.

1. Check batch numbers and condition the kit to +15 °C to +25 °C.
2. Stir Part A 30 s, add Part B (3 : 1 by weight) and mix 1 min; thin 5 – 10 % if required.
3. Apply by roller or squeegee at 0.2 – 0.3 kg/m² per 100 µm.
4. Apply further coats to the specified total.

12. Hold Points & Recoat Windows

Stage	Before starting	Waiting time before next stage	Approved by
1. O-Prime SF	Substrate accepted (Section 9)	Min. 10 h, max. 24 h	QA/QC
2. O-Putty	Previous stage inspected and approved	24 h, then light sanding	QA/QC
3. Freezo Coat	Previous stage inspected and approved	Full cure before cooling the room	QA/QC

13. Curing, Protection & Return to Service

- Protect each layer from rain, dew, dust, water and traffic until it has cured.
- Barricade the area and display warning signs during application and curing.
- Keep ventilation running in enclosed areas until full cure.
- Light foot traffic after 24 hours; full mechanical and chemical load after 7 days at 25 °C (longer at lower temperatures).

14. Quality Control & Inspection Test Plan

Activity	Inspection / Test	Frequency	Record
Material receipt	Product, batch, expiry, packaging condition	Each delivery	Material inspection report
Climate	Air / substrate temperature, RH, dew point (ISO 8502-4)	Start of shift and every 4 h	Daily climate log

Activity	Inspection / Test	Frequency	Record
Substrate	Strength, pull-off (ASTM D7234), moisture (ASTM D4263)	Each area (hold point)	Inspection request
Mixing	Complete kits, ratio, mixing time	Each mix	Daily record
Film thickness	WFT comb gauge (ASTM D4414) during application	Each coat	Daily record
Consumption	Material used vs area	Each stage	Daily record
Adhesion	Pull-off test on trial area / as specified (ASTM D7234 or D4541)	As specified	Test report
Final inspection	Appearance, continuity, finish, details	Each area	Handover checklist

15. Defects & Remedies

Defect	Likely cause	Remedy
Poor adhesion / delamination	Weak, dusty or damp substrate; recoat window exceeded	Remove, re-prepare and reapply; abrade and re-prime if window exceeded
Pinholes / bubbles	Outgassing; air trapped in mixing; thick application	Seal porous areas; mix slowly; spiked roller; sand and recoat
Blushing / tacky surface	High humidity or low temperature during cure	Wash, abrade and recoat in suitable conditions
Soft spots	Incorrect ratio or incomplete mixing	Remove and reapply with complete kits
Colour / shade variation	Different batches; uneven application	Use one batch per area; apply additional coat

16. Health, Safety & Environment

Hazard	Control measures
Skin / eye contact with resins and cement	Gloves, goggles, coveralls; eye-wash station
Inhalation of dust and vapours	Dust extraction; ventilation; masks / respirators
Fire (solvent vapours)	No smoking or hot work; fire extinguisher at the work area
Slips and trips	Housekeeping; barricades and warning signs
Manual handling	Correct lifting; mechanical aids for drums and pails
Electrical equipment	Inspected tools; protected cables with RCD

- Read the Safety Data Sheet of each product before use.
- Keep first-aid kit and eye-wash on site; follow the first-aid measures in each TDS.

16.1 Waste & environment

- Collect empty containers and residues in designated bins.
- Allow small quantities of mixed resin to cure before disposal; dispose of according to local regulations.
- Do not discharge materials or washings into drains or soil.

17. Cleaning

- Clean tools of water-based and cementitious products with water immediately after use.
- Clean tools of epoxy and PU products with the recommended thinner before the material cures.
- Cured material can only be removed mechanically.

18. Limitations

- Do not apply outside the temperature and humidity limits stated in the TDS of each product.
- Do not apply on wet surfaces or when rain or dew is expected before the layer has cured.
- Do not mix or substitute products from other systems without written approval from O MIX.
- Any deviation from this method statement must be approved by the Site Engineer and the consultant.

19. Maintenance

- Clean regularly with neutral detergents and a scrubber-dryer.
- Remove spills promptly, especially chemicals.
- Protect from sharp, heavy point loads and dragging of steel items.
- Re-coat worn traffic areas with the top coat after cleaning and abrading.

Appendix A — Inspection Checklist

Inspection Item	Result	Remarks
Approved submittal and method statement on site	☐ Yes ☐ No	
Mock-up approved	☐ Yes ☐ No	
Substrate tests passed (strength, pull-off, moisture)	☐ Yes ☐ No	
Surface preparation approved	☐ Yes ☐ No	
Stage 1 — O-Prime SF applied and approved	☐ Yes ☐ No	
Stage 2 — O-Putty applied and approved	☐ Yes ☐ No	
Stage 3 — Freezo Coat applied and approved	☐ Yes ☐ No	
Recoat windows respected	☐ Yes ☐ No	
Final tests passed	☐ Yes ☐ No	
Area protected and handed over	☐ Yes ☐ No	

Inspected by	Signature	Date

Appendix B — Daily Application Record

Date	Area / zone	Stage	Batch No.	Qty used	Air / substrate °C	RH %	Sign